



The role of Aldosterone-renin ratio (ALDS) in diagnosing secondary causes of hypertension

**Promo RPP
Combo (code ALDS)
RM380 (tag-on)
RM490 (Single)**

Hypertension is highly prevalent

Malaysian adult population ≈ 22 million



**Prevalence
49.39%**

Hypertensive population ≈ 11 million

Primary Aldosteronism (PA)

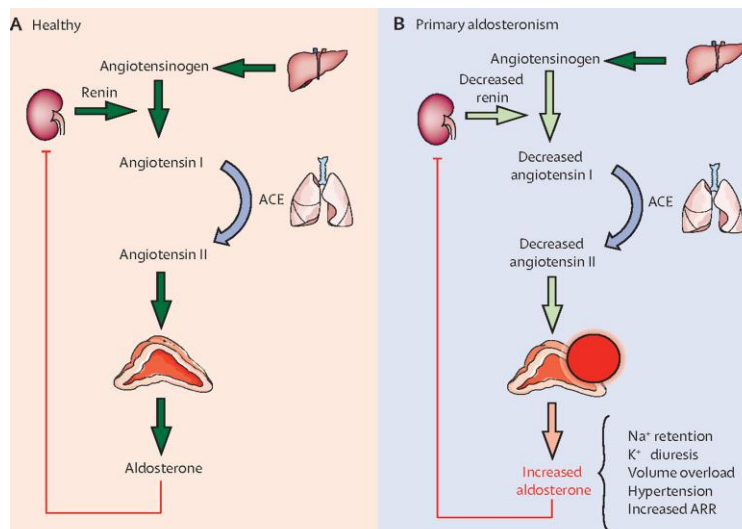
- Most common endocrine cause of hypertension
- Often associated with treatment-resistant hypertension
- Autonomous aldosterone production independent of renin activity, potassium level or volume status
- Strong association with adverse kidney and cardiovascular outcomes
- Prevalence
 - 5-10% among all hypertensive patients
 - Up to 20% in patients with resistant hypertension

Primary Hypertension

- also known as essential hypertension
- no specific cause
- majority of adults are in this category

Secondary Hypertension

- Hypertension caused by an identifiable and potentially reversible condition
- Common underlying causes of secondary hypertension
 - narrowing of the arteries
 - adrenal gland disease
 - hormone abnormalities
 - thyroid abnormalities
 - side effects of some medications
 - obstructive sleep apnea



Why is it important to screen for PA?

Patients with PA have higher cardiovascular morbidity and mortality than patients with essential hypertension
Underreported prevalence among hypertensive patients

Up to 1,000,000,000
People worldwide may have PA

950,000,000
Do not know they do

330,000,000
Could be surgically cured

620,000,000
Could be medically treated

Aldosterone-renin ratio

*most reliable screening test as recommended by
Endocrine Society guidelines for Primary Aldosteronism.*

Positive threshold for PA

Aldosterone-renin ratio ≥ 30 ng/dL per ng/mL/hour

Sample requirement: Renin - 3 ml EDTA, ALD - 5ml Plain
Refer to detail guideline

Who should screen for PA?

- Sustained hypertension, with blood pressure above 150/100 mm Hg on 3 or more occasions
- Hypertension with blood pressure above 140/90 mm Hg despite the use of 3 or more medications
- Controlled blood pressure less than 140/90 mm Hg on 4 or more medications
- Hypertension with spontaneous or diuretic-induced hypokalemia
- Hypertension with obstructive sleep apnea
- Hypertension with an adrenal mass
- Hypertension with a family history of early-onset hypertension or cerebrovascular accident before age 40
- Hypertension with a first-degree relative diagnosed with primary aldosteronism